

4 Enzymes are biological catalysts that have an effect on the energy required for a reaction.

(a) State the meaning of the term **biological catalyst**.

(2)

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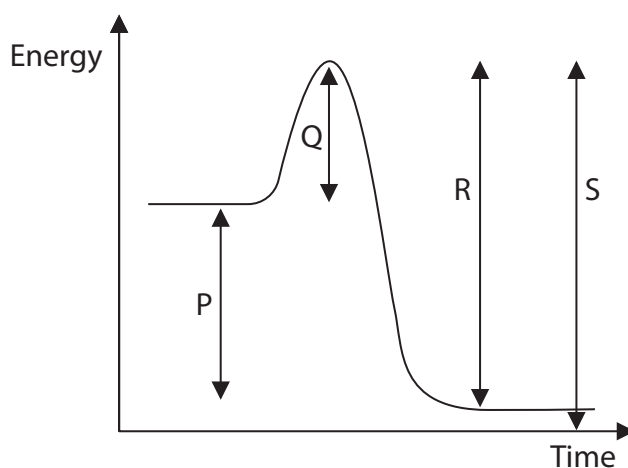
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(b) The graph shows the energy changes during an enzyme catalysed reaction.



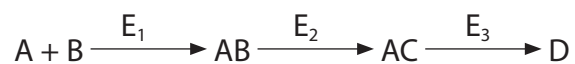
Which arrow shows the energy required to start the reaction?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S



(c) The diagram shows a series of chemical reactions.



Three different enzymes, E_1 , E_2 and E_3 , are involved in these reactions.

Explain why E_1 cannot convert AB into AC or AC into D.

Use the information in the diagram to support your answer.

(3)

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(d) Pectinase is an enzyme that breaks down pectin.

The effect of different combinations of pH and temperature on pectinase activity was investigated.

The volume and concentration of pectinase and pectin were kept the same in all reactions.

The table shows the rate of reaction for each pH and temperature combination tested.

pH	Temperature / °C	Rate of reaction / a.u.
2	5	0
2	30	0
2	60	0
4	5	5
4	30	35
4	60	0
8	5	5
8	30	10
8	60	0

Describe **two** conclusions that can be drawn from these results.

(2)

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(Total for Question 4 = 8 marks)

